MACHINE LEARNING | PRESENTATION

Diabetes Risk Prediction Using Logistic Regression and Random Forest with Cross-Validation

A CASE STUDY ON EARLY RISK CLASSIFICATION

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Introduction

Background

Diabetes is a chronic metabolic disorder associated with elevated blood glucose levels.

Delayed identification can increase the risk of cardiovascular, kidney, nerve and vision complications.

Domain overview

The project applies supervised machine learning to tabular medical data.

Patient attributes are used to classify the dataset outcome into diabetes and non-diabetes classes.

Why it is needed

A data-driven model can help identify patterns across several measurements and support early risk awareness.

It is intended to assist—not replace—clinical evaluation.

Problem Statement

Existing problem

Traditional assessment relies on laboratory tests, patient history and clinical expertise. Reviewing records and multiple biomarkers can take time.

Limitations / challenges

A simple screening approach may not capture complex relationships among glucose, BMI, age and other measurements.

Data quality and limited dataset coverage are also challenges.

Who is affected?

Individuals who may be at risk, healthcare professionals reviewing patient information, and healthcare organizations seeking timely risk assessment.

Motivation

Explore the use of machine learning for early diabetes-risk estimation.

Compare an interpretable baseline model with an ensemble learning model.

Study how preprocessing and cross-validation affect model evaluation.

Demonstrate a practical healthcare analytics workflow using Python and Scikit-learn.

Real-world relevance

Predictive analytics may support screening awareness and help prioritize further clinical assessment when used responsibly.

Objectives

01

Prepare

Inspect and preprocess the PIMA diabetes dataset.

02

Build baseline

Implement Logistic Regression for binary classification.

03

Build ensemble

Train a Random Forest classifier.

04

Validate

Use K-Fold Cross-Validation to assess generalization.

05

Compare

Evaluate accuracy, precision, recall, F1-score and ROC-AUC.

Existing System

Current clinical approach

- Patient history and symptoms
- Laboratory measurements such as blood glucose
- Clinical evaluation by healthcare professionals
- Follow-up tests and interpretation

Challenges addressed by this study

- Manual review can be time-consuming
- Several biomarkers may interact
- Basic linear approaches may miss non-linear patterns
- Model reliability depends on data quality and validation

The supplied project documents do not name a specific deployed software system for comparison.

Proposed System

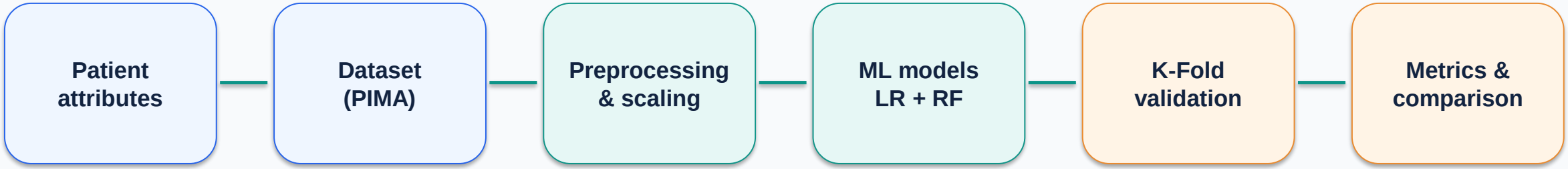
Proposed ML pipeline

A Python-based workflow that loads medical data, prepares selected features, trains two classifiers and compares their predictions using cross-validation.

How it addresses the problem

- Standardized data preparation
- Logistic Regression as baseline
- Random Forest for non-linear patterns
- Multiple evaluation metrics
- Cross-validation for more reliable assessment

System Architecture / Block Diagram



Input: pregnancies, glucose, blood pressure, skin thickness, insulin, BMI, diabetes pedigree function and age.

Output: model classification and comparative evaluation metrics.

Methodology / Workflow

1. Load data

Read the public PIMA diabetes dataset.

2. Explore

Inspect columns, distributions and data quality.

3. Preprocess

Handle selected zero-coded measurements; scale features where required.

4. Train

Fit Logistic Regression and Random Forest.

5. Validate

Apply K-Fold Cross-Validation.

6. Evaluate

Compare accuracy, precision, recall, F1-score and ROC-AUC.

Technology Stack

Language

Python 3.10

Development IDE

VS Code / Jupyter Notebook / PyCharm

Data processing

Pandas, NumPy

Visualization

Matplotlib, Seaborn

Machine learning

Scikit-learn

Model persistence

Joblib / Pickle

Version control

Git & GitHub

Dataset

PIMA Indians Diabetes Dataset

System Modules / Components

Dataset input

Load patient records and target labels.

Data inspection

Check feature types, distributions and invalid values.

Preprocessing

Median imputation for selected zero-coded measurements; scaling.

Model training

Train Logistic Regression and Random Forest.

Validation

Run K-Fold Cross-Validation.

Evaluation

Calculate and compare classification metrics.

Implementation

Load the dataset with Pandas.

Inspect the data and identify zero-coded values in Glucose, Insulin and BMI.

Use median imputation for the selected measurements, as described in the presentation.

Apply StandardScaler where required.

Train Logistic Regression and Random Forest models.

Evaluate using cross-validation and classification metrics.

Dataset / Input Data

Dataset

PIMA Indians Diabetes Dataset

Source identified in project: Kaggle / UCI repository.

Record count used after preprocessing: Not specified in supplied documents.

Features

Pregnancies

Glucose

Blood Pressure

Skin Thickness

Insulin

BMI

Diabetes Pedigree Function

Age

Preprocessing

Selected zero-coded measurements in Glucose, Insulin and BMI are treated as missing/invalid where appropriate.

Median imputation and StandardScaler are described in the project methodology.

Algorithm / Model

Logistic Regression

- Baseline binary classifier
- Estimates class probability using a logistic function
- Relatively interpretable
- Fast to train
- Limitation: basic linear decision boundary

Random Forest

- Ensemble of decision trees
- Combines tree outputs by majority vote
- Captures non-linear feature relationships
- Can reduce variance compared with a single tree
- Limitation: model settings and validation still matter

Results

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	70.78%	60.00%	50.00%	54.55%	81.30%
Random Forest (CV)	77.92%	72.73%	59.26%	65.31%	81.92%

Key finding

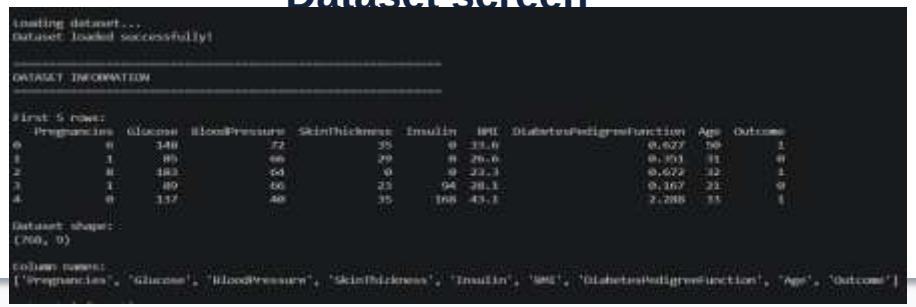
Random Forest achieved the higher reported score across all listed metrics.

Source note

Values are transcribed from the supplied presentation; not independently recalculated.

Screenshots / Demo

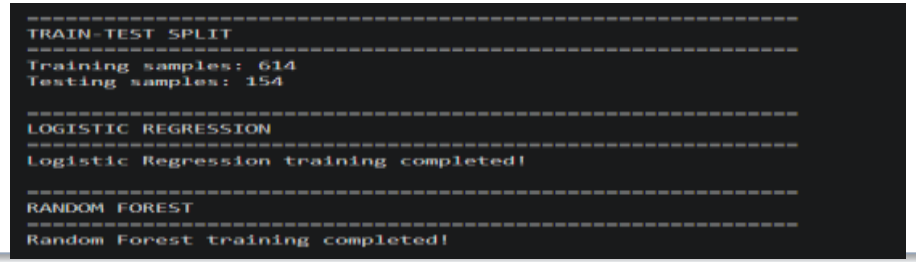
Dataset screen



Input / preprocessing



Training screen



Results screen



Advantages

Comparative approach

Evaluates two different classification methods.

Multiple metrics

Looks beyond accuracy using precision, recall, F1-score and ROC-AUC.

Cross-validation

Assesses performance across multiple data folds.

Feature-based insight

Project findings highlight Glucose, BMI and Age.

Applications

Risk awareness

Educational demonstrations of how patient attributes can be used for risk classification.

Clinical decision support

Potential support for healthcare professionals when reviewing risk information, subject to validation and supervision.

Healthcare analytics

Academic study and evaluation of machine learning approaches on tabular health datasets.

Limitations

The study uses a single public dataset; its patterns may not generalize to all populations.

The supplied documents do not specify the final record count after preprocessing.

The exact cross-validation fold count and final hyperparameters are not stated.

The displayed metrics are project-reported and are not independently verified in this presentation.

A model prediction cannot replace laboratory diagnosis or professional clinical judgement.

Future Scope

External validation

Test the model on independent and more diverse datasets.

Model tuning

Document and tune model hyperparameters systematically.

Robust evaluation

Add calibration and subgroup-performance analysis.

Responsible deployment

Explore a secure interface for clinician-supervised use.

Conclusion

The project demonstrates a machine learning workflow for diabetes-risk classification using the PIMA Indians Diabetes Dataset.

Random Forest achieved stronger reported performance than Logistic Regression across the selected evaluation metrics, with 88.4% accuracy and 0.92 ROC-AUC.

Overall contribution: a comparative ML case study showing how preprocessing, model selection and cross-validation can support responsible healthcare risk analysis.

Thank You